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sps: Sequential Poisson sampling in R

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Software

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Summary

Sequential Poisson sampling is a simple and flexible way to draw a probability-proportional-to-size sample without replacement that's often useful in practice (Ohlsson, 1998). The sps package gives a complete implementation of the sequential Poisson method for drawing random samples with R (R Core Team, 2026). Unlike other software in this space, sps is focused on tools related to the sequential Poisson method—it offers a simple interface for this sampling method along with complementary functionality that is useful in practice and not broadly available elsewhere.

Statement of need

Drawing random samples with unequal probabilities of selection is an important step for many statistical projects (e.g., conducting a sample survey, auditing and evaluation). Sequential Poisson sampling is a method for drawing a random sample from a finite population of units without replacement, with each unit sampled proportional to some measure of size (e.g., revenue for drawing a sample of businesses). It can be motivated as a variation of the ordinary Poisson method that always draws a fixed number of units (Ohlsson, 1998), a natural extension of simple random sampling without replacement, and it belongs to the more general family of order-sampling methods (Rosén, 1997). Although there are many methods for unequal-probability sampling (e.g., Tillé, 2020, Chapter 5), the sequential Poisson method is notable for being a simple, fast, and flexible method, with many nice properties not shared by other methods—consequently, it is often useful in practice. In particular, this method supports coordinating samples and reproducibly drawing samples with permanent random numbers, does not require the sample size to be fixed in advanced (i.e., the sample can be draw iteratively), and can work with techniques for ordinary Poisson sampling to the extent that sequential Poisson sampling offers a fixed-size approximation to ordinary Poisson sampling.

State of the field

There are many packages on CRAN for drawing random samples, but these generally do not include the sequential Poisson method (e.g., Afifanto, 2024; Gambino, 2021; Rojas, 2020). The `sampling` package (Tillé & Matei, 2025) is probably the best known R package for drawing samples and contains functions for many different methods, including the sequential Poisson method. The focus of the `sampling` package, however, is to provide a unified interface to many different sampling methods—consequently, it lacks specific features that are relevant for the sequential Poisson method and notably does not allow for stratification, take-all units, or the use of permanent random numbers. By contrast, the `prnsamplr` package (Coder Gylling, 2025) allows for the use of stratification and permanent random numbers with Pareto order sampling, but does not feature other order-sampling methods (like sequential Poisson). The `sondage` package (Dicko, 2026) is relatively new and improves upon `sampling`, but is again

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a generalist sampling package focused on providing many different (unstratified) sampling algorithms and lacks features that are specific to the sequential Poisson method.

The sps package fills a gap in existing software for drawing random samples by providing a complete implementation of the sequential Poisson method that is designed to be used in practice. It's built around features that distinguish the sequential Poisson method from other methods, making it well-suited as a stand-alone package rather than an extension to an existing package.

Software design

The sps package is designed to provide a complete implementation of the sequential Poisson sampling method that is useful for typical sampling problems. The design closely follows the theory by Ohlsson (1998) and the generalization to other order-sampling schemes by Rosén (1997). The package interface is influenced by the pps package (Gambino, 2021), as it provides a simple and minimal model for stratified proportional-to-size sampling. Compared to existing tools, there is a greater focus on take-all units that are always included in the sample as these units are important in practice.

One implication of keeping the design close to the theory is the use of partial sorting algorithms to improve performance, as all order sampling methods work by partially sorting an auxiliary variable to draw a sample. The `kit` package (Jacob, 2026) is used for this—it provides a stable partial-sorting method that works well when drawing a relatively small sample from a large population, while ensuring that partial sorting agrees with a (stable) full sort. Another implications of working close to Ohlsson (1998)'s paper is the use of a novel method for efficiently calculating inclusion probabilities to deal with a potential bias of the sequential Poisson method.

Despite the focus on a single sampling method, `sps` is designed to be used for practical applications—it contains various complementary functions to draw a sample so that it can work on its own, including methods for all other order-sampling methods and ordinary Poisson sampling, all while being broadly interoperable with other packages in the R ecosystem. For example, the `sps::prop_allocation()` function provides many methods for constructing a proportional-to-size allocation for a stratified sample. Although a common approach, this is one of many different ways to allocate a sample size across strata, and the `sps::sps()` function to draw a sample accepts any (feasible) integer vector as an allocation to work with other R packages (e.g., `optimall`, Yang et al., 2025).

Research impact statement

The sps package is broadly available to researchers through the usual channels for statistical software and it has been available on CRAN since 2021. sps is featured in the list of Awesome official statistics software (Bosch, 2022) and is part of the CRAN Task View for Official Statistics & Survey Statistics (Templ et al., 2025). It has also been used for several project at Statistics Canada where the sequential Poisson method is often used for drawing samples of businesses (e.g., Batubenga & Kirby-McGregor, 2025).

Example usage

There are several vignettes, along with detailed help pages, explaining how to use the sps package. Despite extensive documentation of the package, it is nonetheless useful to show a small example to see the basics of sps and motivate some of the design choices.

To start, consider a frame with an enumeration of all businesses operating in a region, along with their revenue for a given year. The goal is to draw a sample of businesses proportional

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to revenue so that businesses with a larger revenue are more likely to be included in the sample. Importantly, each business has a *permanent random number* that makes it possible to reproducibly draw a sample.

```
set.seed(45499)

frame <- data.frame(
  business = sprintf("B%03d", 1:1e3),
  revenue = round(rlnorm(1e3) * 1000),
  prn = runif(1e3)
)

head(frame)
```

```
#>   business revenue       prn
#> 1     B001     770 0.7613567
#> 2     B002     262 0.8288327
#> 3     B003     590 0.7873136
#> 4     B004     266 0.2360220
#> 5     B005     275 0.9241041
#> 6     B006    2117 0.1985771
```

It is now easy to a draw a random sample using the sequential Poisson method.

```
samp <- with(frame, sps::sps(revenue, n = 100, prn = prn))
head(samp)
```

```
#> [1]  9 22 24 27 33 48
```

The result is an integer vector that can be used to select which businesses from the frame are in the sample, along with the design weight for each selected unit.

```
frame[samp, ] |>
  transform(design_weight = weights(samp)) |>
  head()
```

```
#>    business revenue         prn design_weight
#> 9      B009    1249 0.048897737     12.958577
#> 22     B022    1756 0.064713425      9.217120
#> 24     B024    1576 0.005890128     10.269837
#> 27     B027    1736 0.030081901      9.323308
#> 33     B033    1320 0.019433044     12.261563
#> 48     B048     881 0.025482835     18.371467
```

An equivalent way to draw this sample is to make an iterator that draws units one at a time without changing the set of previously sampled units.

```
sps_iter <- with(frame, sps::sps_iterator(revenue, prn = prn))
samp2 <- integer(0)

for (. in 1:100) samp2 <- c(sps_iter(), samp2)

setequal(samp, samp2)
```

```
#> [1] TRUE
```

Once the number of units in the sample is known, inclusion probabilities can be calculated for each unit on the frame to get the design weight for each unit in the iteratively-drawn sample.

```
pik <- sps::inclusion_prob(frame$revenue, n = 100)
setequal(weights(samp), 1 / pik[samp2])
```

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```
#> [1] TRUE
```

This feature of sequential Poisson sampling is not shared by other unequal-probability methods, but is useful for topping up a sample or drawing a sample without knowing the fixed number of units to be included in the sample (e.g., as with field work).

AI usage disclosure

No generative AI tools were used to make any part of sps or this paper.

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